

Appendix: Learning and Free-Riding in International Climate Policymaking

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A Formal Proofs

Proof of Proposition 1. To solve for the equilibrium, I conjecture the existence of a strategy for country 1 $a_1 = \alpha_1(x_1)$ and assume that $\alpha_1(x_1)$ is one-to-one. Proceeding by backward induction first consider country 2's effort investment given her signal x_2 and country 1's effort a_1 . Given that $\alpha_1(x_1)$ is one-to-one, we have $x_1 = \alpha_1^{-1}(a_1)$ and so country 2's posterior update about θ is $\theta|x_2, a_1 \sim N(\frac{\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2}{\gamma + 2\beta}, \frac{1}{\gamma + 2\beta})$. Country 2 therefore solves

$$\max_{a_2} E[\theta|x_2, a_1]g(a_1 + \lambda a_2) - c_2(a_2).$$

Differentiating with respect to a_2 , country 2's first-order condition is

$$E[\theta|x_2, a_1]g'(a_1 + \lambda a_2)\lambda - c'_2(a_2) = 0.$$

Note that if $E[\theta|x_2, a_1] < 0$, which occurs when $x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1)$ then $E[\theta|x_2, a_1] < 0$ and country 2 exerts effort in the opposite direction. Given the functional form specifications and solving the above first-order condition, we have the following piecewise function:

$$\alpha_2(x_2, a_1) = \begin{cases} -\frac{1}{\lambda}a_1 + \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & x_2 \geq -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1) \\ -\frac{1}{\lambda}a_1 - \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1) \end{cases}$$

It is clear from the second-order condition that for any x_2 and any a_1 , $\alpha_2(x_2, a_1)$ is unique as the second-order condition is always negative:

$$E[\theta|x_2, a_1]g''(a_1 + \lambda a_2)\lambda^2 - c''_2(a_2) < 0.$$

Now consider country 1's effort choice. Given his own signal x_1 , he believes that $\theta|x_1 \sim N(\frac{\gamma\mu + \beta x_1}{\gamma + \beta}, \frac{1}{\gamma + \beta})$ and that country 2's signal $x_2|x_1 \sim N(\frac{\gamma\mu + \beta x_1}{\gamma + \beta}, \frac{2\beta + \gamma}{\beta(\gamma + \beta)})$. Let $m = \frac{\gamma\mu + \beta x_1}{\gamma + \beta}$ and $z = \sqrt{\frac{\beta(\gamma + \beta)}{2\beta + \gamma}}$. Further, denote $q = \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2}$ and $t = -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1)$. By backward induction, country 1's problem is to maximize

$$\max_{a_1} \int_{-\infty}^t [mg(-q)z\phi(z(x_2 - m))] dx_2 + \int_t^{\infty} [mg(q)z\phi(z(x_2 - m))] dx_2 - c_1(a_1).$$

Differentiating with respect to a_1 , country 1's first-order condition is

$$\begin{aligned} FOC &= mg(0)z\phi(z(t - m))\frac{dt}{da_1} + \int_{-\infty}^t -mg'(-q)\frac{dq}{da_1}z\phi(z(x_2 - m)) dx_2 \\ &\quad - mg(0)z\phi(z(t - m))\frac{dt}{da_1} + \int_t^{\infty} mg'(q)\frac{dq}{da_1}z\phi(z(x_2 - m)) dx_2 - c_1 = 0 \\ &= \int_{-\infty}^t \frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{1}{\alpha'_1(\alpha_1^{-1}(a_1))} z\phi(z(x_2 - m)) dx_2 \end{aligned}$$

$$\begin{aligned}
& + \int_t^\infty \frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{1}{\alpha_1'(\alpha_1^{-1}(a_1))} z\phi(z(x_2 - m)) dx_2 - c_1 = 0 \\
\Leftrightarrow \quad \alpha_1'(\alpha_1^{-1}(a_1)) &= \frac{(\gamma\mu + \beta x_1)\beta\lambda}{2c_2c_1(2\beta + \gamma)(\gamma + \beta)}.
\end{aligned}$$

It is clear that since q is increasing in $\alpha_1^{-1}(a_1)$, country 1's optimal strategy does not contain any "flat spots" as it is always optimal for him to induce greater effort from country 2. Observe that for $x_1 < -\frac{\mu\gamma}{\beta}$, $m < 0$ and so country 1 would then exert effort in the negative direction. By equilibrium conjecture, $a_1 = \alpha_1(x_1)$ so $\alpha_1^{-1}(a_1) = x_1$ and integrating with respect to x_1 yields

$$\alpha_1(x_1) = \frac{\beta\lambda x_1(2\gamma\mu + \beta x_1)}{4c_1c_2(2\beta^2 + 3\beta\gamma + \gamma^2)} + C.$$

The constant of integration is pinned down by the boundary condition that, at $x_1 = -\frac{\gamma\mu}{\beta}$, we have $E[\theta|x_1] = 0$. The equilibrium effort is thus

$$\alpha_1(x_1) = \begin{cases} \frac{\lambda(\gamma\mu + \beta x_1)^2}{4c_1c_2(2\beta + \gamma)(\beta + \gamma)} & x_1 \geq -\frac{\gamma\mu}{\beta} \\ \frac{-\lambda(\gamma\mu + \beta x_1)^2}{4c_1c_2(2\beta + \gamma)(\beta + \gamma)} & x_1 < -\frac{\gamma\mu}{\beta}. \end{cases}$$

Note that this is one-to-one in x_1 , confirming that $\alpha_1(x_1)$ is one-to-one in equilibrium. This means that $\alpha_1^{*-1}(\cdot)$ is well-defined so country 2 knows $x_1 = \alpha_1^{*-1}(a_1)$ in equilibrium.

Finally, observe that the second order condition is

$$-\frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{\alpha_1''(\alpha_1^{-1}(a_1))}{(\alpha_1'(\alpha_1^{-1}(a_1)))^3} < 0,$$

so the solution $\alpha_1(x_1)$ is the unique maximizer of country 1's utility. □

Proof of Corollary 1. Immediate given the equilibrium strategy of country 1:

$$\begin{aligned}
\frac{d\alpha_1(x_1)}{dx_1} &= \frac{\beta\lambda|\gamma\mu + \beta x_1|}{2c_1c_2(2\beta^2 + 3\beta\gamma + \gamma^2)} \geq 0. \\
\frac{dE[\theta|x_2, a_1]}{da_1} &= \frac{\beta}{2\beta + \gamma} \frac{1}{\frac{d\alpha_1(x_1)}{dx_1}} \geq 0.
\end{aligned}$$

Note that the absolute value in $\frac{d\alpha_1(x_1)}{dx_1}$ is simply because if $\gamma\mu + \beta x_1 < 0$, then we are on the lower branch of $\alpha_1(x_1)$ and the negative signs cancel. □

Proof of Corollary 2. Given country 2's equilibrium effort function as defined in Equation 1,

$$\frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} = -\frac{1}{\lambda} < 0.$$

$$\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} = \frac{\lambda \beta |\gamma \mu + \beta \alpha_1^{-1}(a_1) + \beta x_2|}{2c_2^2(2\beta + \gamma)^2} > 0.$$

Note that the absolute value in $\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)}$ comes from the piecewise nature of $\alpha_2(x_2, a_1)$, recall that if $x_2 < -\frac{\gamma \mu}{\beta} - \alpha_1^{-1}(a_1)$ then $E[\theta|x_2, a_1] < 0$. $\square$

Proof of Proposition 2. The total derivative of $\alpha_2(x_2, a_1)$ with respect to a_1 is

$$\frac{d\alpha_2(x_2, a_1)}{da_1} = \frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} + \frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} \cdot \frac{d\alpha_1^{-1}(a_1)}{da_1}.$$

From Corollary 2, we know that $\frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} = -\frac{1}{\lambda} < 0$ for any x_2 . Similarly, $\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} = \frac{\lambda \beta |\gamma \mu + \beta \alpha_1^{-1}(a_1) + \beta x_2|}{2c_2^2(2\beta + \gamma)^2} > 0$. By the inverse function theorem, $\frac{d\alpha_1^{-1}(a_1)}{da_1} = \frac{1}{\alpha_1'(\alpha_1^{-1}(a_1))}$, and from the derivation of country 1's optimal effort, we know $\alpha_1'(\alpha_1^{-1}(a_1)) = \frac{\beta \lambda |\gamma \mu + \beta \alpha_1^{-1}(a_1)|}{2c_2 c_1(2\beta + \gamma)(\gamma + \beta)}$. Imposing $\alpha_1^{-1}(a_1) = x_1$ and simplifying yields

$$\frac{d\alpha_2(x_2, a_1)}{da_1} = -\frac{1}{\lambda} + \frac{|\gamma \mu + \beta x_1 + \beta x_2|}{c_2(2\beta + \gamma)} \cdot \frac{c_1(\beta + \gamma)}{|\gamma \mu + \beta x_1|}.$$

Hence $\frac{d\alpha_2(x_2, a_1)}{da_1} > 0$ iff

$$\frac{|\gamma \mu + \beta x_1 + \beta x_2|}{|\gamma \mu + \beta x_1|} > \frac{c_2(2\beta + \gamma)}{\lambda c_1(\beta + \gamma)},$$

or

$$\frac{|E[\theta|x_2, a_1]|}{|E[\theta|x_1]|} > \frac{c_2}{\lambda c_1}.$$

$\square$

B Cross-Country Climate Actions: Robustness and Ancillary Analysis

Statistic	N	Mean	St. Dev.	Min	Max
Climate Laws (Count)	7,200	0.643	1.003	0.000	4.937
Climate Laws (Binary)	7,200	0.713	1.014	0.000	4.305
Other Laws	7,200	191.593	206.168	5	832
EPS	1,240	2.043	1.215	0.000	4.998
Avg. Other EPS	1,612	1.780	0.809	0.590	2.942
CAPMF	1,700	1.252	1.009	0.000	3.860
Avg. Other CAPMF	1,762	2.184	1.688	0.104	4.956

Table A.1: Summary Statistics for Climate Laws and Climate Policy Stringency

The main results employ country $\times$ year time trends. Table A.2 shows robustness of the findings without any temporal controls.

	Laws (Count)		Laws (Binary)		EPS		CAPMF	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Other Laws)	0.530*** (0.012)	0.819*** (0.049)	0.526*** (0.011)	0.867*** (0.053)				
Avg. Other Stringency					0.918*** (0.013)	0.706*** (0.143)	0.972*** (0.003)	0.818*** (0.051)
Observations	7,000	7,000	7,000	7,000	1,200	1,200	1,650	1,650
R ²	0.302	0.331	0.314	0.344	0.900	0.904	0.960	0.962
Within R ²	0.277	0.307	0.273	0.305	0.845	0.852	0.958	0.960
DV Mean	0.643	0.643	0.713	0.713	2.04	2.04	1.25	1.25
Number of Countries	200	200	200	200	40	40	50	50
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Country $\times$ Year trends		✓		✓		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.2: Effects of Previous Law Adoption and Policy Stringency on Climate Policymaking (with and without Time Trends)

The main results include a country fixed effect to parse out any time-invariant factors that lead countries to adopt laws or pursue environmental policies in a heterogeneous fashion. However, in so doing, the regression coefficient β thus targets the within-country correlation of the effect of other nations' climate actions on the adoption of climate laws. Table A.3 removes country fixed effects and shows that the results still hold.

	Laws (Count)		Laws (Binary)		EPS		CAPMF	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Other Laws)	0.530*** (0.012)	0.803*** (0.048)	0.525*** (0.012)	0.849*** (0.052)				
Avg. Other Stringency					0.904*** (0.013)	0.409** (0.152)	0.971*** (0.003)	0.801*** (0.051)
Observations	7,000	7,000	7,000	7,000	1,200	1,200	1,650	1,650
R ²	0.267	0.270	0.257	0.262	0.529	0.536	0.909	0.911
Adjusted R ²	0.267	0.270	0.257	0.262	0.528	0.536	0.909	0.910
DV Mean	0.643	0.643	0.713	0.713	2.04	2.04	1.25	1.25
Year trends		✓		✓		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.3: Effects of Previous Law and Policy Stringency on Climate Policymaking (without Country Fixed Effects)

B.1 Climate Laws

In the main text, I employ country $\times$ year time trends to capture the effects of potential secular increases in the demand for climate policy. Table A.4 shows that results are robust to different types of time trends. Specifically, I estimate linear, quadratic, and cubic yearly time trends. I also estimate year random effects.

	Laws (Count)			Laws (Binary)		
	(1)	(2)	(3)	(4)	(5)	(6)
log(Other Laws)	0.819*** (0.048)	0.968*** (0.047)	0.388*** (0.064)	0.867*** (0.052)	0.911*** (0.052)	0.315*** (0.064)
Observations	7,000	7,000	7,000	7,000	7,000	7,000
R ²	0.305	0.310	0.330	0.319	0.319	0.340
Within R ²	0.280	0.285	0.306	0.278	0.279	0.301
Country fixed effects	✓	✓	✓	✓	✓	✓
Time trends	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.4: Effects of Climate Laws with Alternate Time Trends

	Laws (Count)	Laws (Binary)
	(1)	(2)
(Intercept)	−1.019*** (0.213)	−0.963*** (0.192)
log(Other Laws)	0.530*** (0.050)	0.526*** (0.040)
Country fixed effects	✓	✓
AIC	17652.524	17938.139
BIC	19043.818	19329.433
Log Likelihood	−8623.262	−8766.069
Num. obs.	7000	7000
Num. groups: year	35	35
Var: year (Intercept)	0.082	0.052
Var: Residual	0.655	0.685

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A.5: Effects of Climate Laws with Year Random Effects

While the results between other nations' laws and the tendency to adopt laws should be treated as descriptive or correlational, I also estimate these models with the inclusion of some time-varying controls. These control variables are meant to parse time-varying variation away from nations' climate policymaking behavior. Specifically, I estimate the revised model

$$\text{Laws}_{i,t} = \beta \log(\text{Other Laws}_{-i,t-1}) + X'_{i,t-1}\gamma + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

where the term $X'_{i,t-1}\gamma$ captures these controls. I include a lagged dependent variable, GDP per capita, population, and a country's growth rate (all from the World Bank), and the size of a country's winning coalition to proxy for regime type ([Bueno de Mesquita and Smith 2022](#)). Results are shown in Table A.6.

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.557*** (0.033)	0.005 (0.076)	0.544*** (0.035)	0.224*** (0.079)
Lagged DV	0.071*** (0.019)	0.023 (0.019)	0.041** (0.016)	0.003 (0.017)
log(GDP per capita)	0.148 (0.101)	-0.097 (0.196)	0.168 (0.104)	0.027 (0.227)
log(Population)	0.089 (0.092)	-0.246 (0.655)	0.141 (0.105)	0.154 (0.552)
Growth	0.004 (0.003)	0.002 (0.003)	0.005* (0.003)	0.003 (0.004)
Winning Coalition Size	-0.419** (0.174)	-0.766*** (0.249)	-0.242 (0.183)	-0.447* (0.244)
Observations	5,075	5,075	5,075	5,075
R ²	0.366	0.396	0.366	0.389
Within R ²	0.345	0.376	0.331	0.355
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Controls	✓	✓	✓	✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.6: Effects of Climate Laws with Controls

We may be worried that only richer countries or only countries that are emissions-intensive need to pass laws. Tables A.7 and A.8 weight results by countries' GDP per capita and emissions per capita.

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.613*** (0.011)	0.004 (0.076)	0.597*** (0.011)	0.248*** (0.077)
Observations	5,863	5,863	5,863	5,863
R ²	0.350	0.381	0.355	0.379
Within R ²	0.327	0.360	0.318	0.343
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GDP per capita	GDP per capita	GDP per capita	GDP per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.7: Effects of Climate Laws (Weighted by GDP per capita)

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.506*** (0.031)	0.677*** (0.100)	0.498*** (0.031)	0.752*** (0.097)
Observations	6,545	6,545	6,545	6,545
R ²	0.293	0.329	0.301	0.335
Within R ²	0.257	0.295	0.245	0.282
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GHG per capita	GHG per capita	GHG per capita	GHG per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.8: Effects of Climate Laws (Weighted by Emissions per capita)

B.2 Policy Stringency

In the main text, I employ country $\times$ year time trends to capture the effects of potential secular increases in the demand for climate policy. Table A.9 shows that results are robust to different types of time trends. Specifically, I estimate linear, quadratic, and cubic yearly time trends. I also estimate year random effects.

	EPS			CAPMF		
	(1)	(2)	(3)	(4)	(5)	(6)
Average Other Stringency	0.705*** (0.141)	0.698*** (0.137)	0.339 (0.224)	0.818*** (0.051)	0.823*** (0.055)	0.646*** (0.047)
Observations	1,200	1,200	1,200	1,650	1,650	1,650
R ²	0.901	0.901	0.902	0.962	0.962	0.963
Within R ²	0.847	0.847	0.848	0.960	0.960	0.961
Country fixed effects	✓	✓	✓	✓	✓	✓
Time trends	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.9: Effects of Policy Stringency with Alternate Time Trends

	EPS	CAPMF
	(1)	(2)
(Intercept)	−0.546*** (0.078)	−0.327*** (0.041)
Average Other Stringency	0.916*** (0.015)	0.966*** (0.017)
Country fixed effects	✓	✓
AIC	1320.345	−540.609
BIC	1539.218	−253.957
Log Likelihood	−617.172	323.305
Num. obs.	1200	1650
Num. groups: year	30	33
Var: year (Intercept)	0.003	0.008
Var: Residual	0.148	0.033

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A.10: Effects of Policy Stringency with Year Random Effects

While the results between other nations' stringency and the country i 's policy stringency should be treated as descriptive or correlational, I also estimate these models with the inclusion of some time-varying controls. These control variables are meant to parse time-varying variation away from nations' climate policymaking behavior. Specifically, I estimate the revised model

$$\text{Stringency}_{i,t} = \beta \text{ Average Stringency}_{-i,t-1} + X'_{i,t-1}\gamma + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

where the term $X'_{i,t-1}\gamma$ captures these controls. I include a lagged dependent variable, GDP per capita, population, and a country's growth rate (all from the World Bank), and the size of a country's winning coalition to proxy for regime type ([Bueno de Mesquita and Smith 2022](#)). Results are shown in Table A.11.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.184*** (0.031)	0.134** (0.056)	0.213*** (0.034)	0.080*** (0.028)
Lagged DV	0.745*** (0.024)	0.738*** (0.024)	0.767*** (0.034)	0.753*** (0.036)
log(GDP per capita)	0.158*** (0.053)	0.529* (0.262)	0.107*** (0.025)	-0.097 (0.090)
log(Population)	0.238* (0.125)	-0.389 (0.924)	0.143*** (0.037)	-0.873* (0.438)
Growth	-0.006* (0.003)	-0.005 (0.003)	0.002 (0.001)	-0.0002 (0.001)
Winning Coalition Size	0.057 (0.242)	0.116 (0.510)	0.045 (0.091)	-0.234 (0.145)
Observations	1,165	1,165	1,469	1,469
R ²	0.952	0.954	0.977	0.978
Within R ²	0.927	0.930	0.975	0.977
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Controls	✓	✓	✓	✓

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.11: Effects of Policy Stringency with Controls

We may be worried that only richer countries or only countries that are emissions-intensive have greater policy stringency. Tables A.12 and A.13 weight results by countries' GDP per capita and emissions per capita.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.923*** (0.013)	0.732*** (0.137)	0.968*** (0.005)	0.796*** (0.052)
Observations	1,170	1,170	1,523	1,523
R ²	0.898	0.902	0.958	0.961
Within R ²	0.845	0.852	0.956	0.959
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GDP per capita	GDP per capita	GDP per capita	GDP per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.12: Effects of Policy Stringency (Weighted by GDP per capita)

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.927*** (0.011)	0.785*** (0.131)	0.972*** (0.004)	0.782*** (0.052)
Observations	1,200	1,200	1,617	1,617
R ²	0.903	0.907	0.958	0.960
Within R ²	0.854	0.859	0.956	0.959
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GHG per capita	GHG per capita	GHG per capita	GHG per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.13: Effects of Policy Stringency (Weighted by Emissions per capita)

Table A.14 shows robustness to measuring the behavior of other nations at the median rather than the average policy stringency, which eases concerns about extreme values.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Median Stringency	0.919*** (0.013)	0.746*** (0.168)	0.972*** (0.003)	0.877*** (0.051)
Observations	1,200	1,200	1,650	1,650
R ²	0.900	0.904	0.962	0.963
Within R ²	0.844	0.850	0.960	0.961
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.14: Effects of Median Policy Stringency

B.3 U.S. Presidential Elections and Global Climate Action

As an additional test, I examine the effects of United States presidential elections on the adoption of climate laws. As the leader of one of the globe’s largest emitters, the identity of the American president heavily skews the trajectory of climate policy worldwide: in 2016, Donald Trump campaigned on withdrawing the United States from the Paris Agreement and retrenching the U.S. from the global climate regime, while Joe Biden’s 2020 campaign promoted American engagement in climate policymaking. The outcome of the 2016 (2020) election thus represents a negative (positive) perturbation to other countries’ best responses and subsequently informs their optimal decisionmaking in light of expected American behavior following the election. The classical logic predicts that other nations should increase (decrease) their adoption of climate laws following Trump’s (Biden’s) victory.

To capture the effects of the elections, I run a regression discontinuity in time at the country-month level using the following local linear regression specification:

$$\text{Laws}_{i,m} = \beta \mathbb{1}(m > t_k) + \varphi f(m - t_k) + \alpha_i + \varepsilon_{i,m}.$$

The dependent variable is either, as in the first empirical test, a binary indicator for whether country i adopted a climate law in month m or the logged count of climate laws adopted by country i in month m . As the running variable is time, the independent variable of interest is an indicator of whether time at month m has exceeded t_k , the calendar date of the election (either November 8 2016 or November 3 2020), and $f(\cdot)$ is a smooth function of time, which is measured in days until the election. Fixed effects at the country level α_i are also included to capture country-specific idiosyncrasies in law adoption. I report bias-corrected estimates with optimal bandwidth selection, weight observations using a triangular kernel (Calonico, Cattaneo and Titiunik 2014), and cluster standard errors by country.

The sign of β identifies the sign of the local average treatment effect of the U.S. election on other nations’ adoption of climate laws right at the date of the election.¹ Accordingly, this estimate should be thought of as countries’ expectations of the value of future effort given the winner of the election. A free-riding argument would expect the sign of β to be positive (negative) in 2016 (2020). In this account, Trump’s apathy toward climate policies increases the need for other nations to compensate, while Biden’s willingness to act decreases the incentives for others to exert costly effort.

Figure A.1 presents graphical evidence of a change in countries’ propensities to adopt climate laws around the 2016 and 2020 United States elections.² I fit a linear polynomial as well as a loess smoother on each side of the election cutoff. The left panel of the figure suggests that there is a small reduction in the adoption of climate laws when Donald Trump wins the election in 2016. By contrast, the right panel documents a large positive surge in

¹Given the design, bias in the LATE would emerge if countries strategically adopt climate laws before or after the election. For example, in 2016, the relevant bias in favor of the free-riding argument is if countries were to delay adoption of climate laws until after the election, attenuating the LATE. However, such a case seems unlikely as adoption would only become more difficult with a U.S. president who is less favorable to climate action in power.

²RD plots use the logged count outcome; plots for the binary dependent variable display similar results.

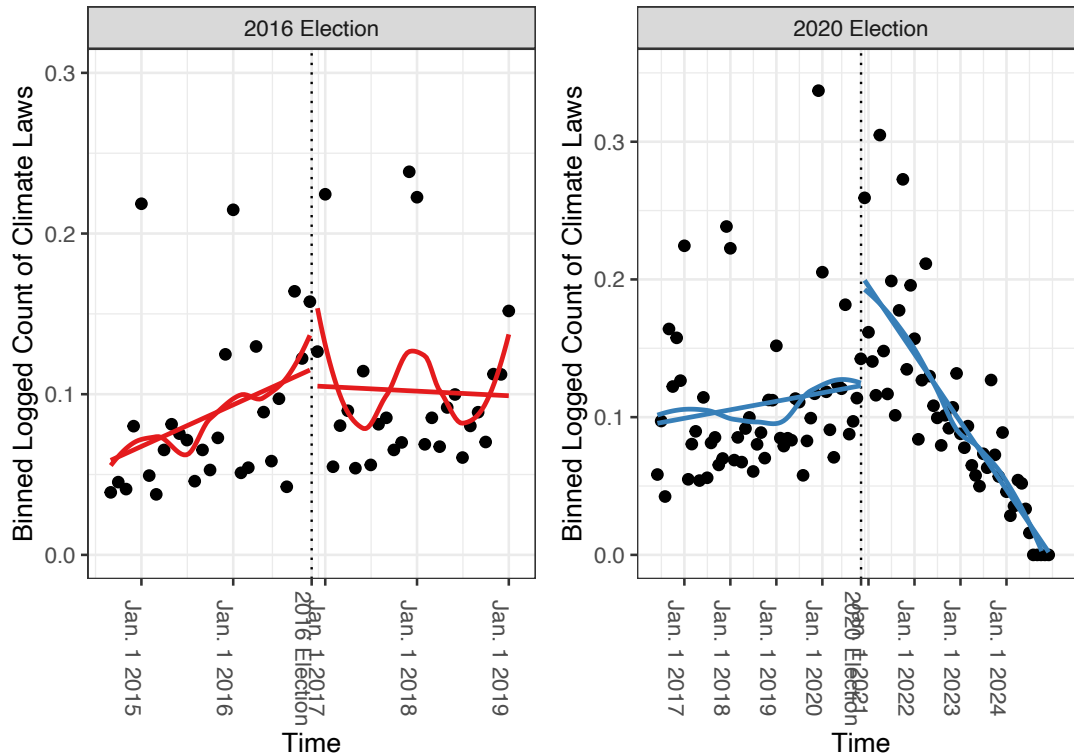


Figure A.1: RD Plots of U.S. Presidential Elections on Climate Law Adoption

the number of laws adopted when Joe Biden wins the election in 2020.

Table A.15 formalizes these estimates, documenting a negative, but statistically indistinguishable from zero, effect in 2016, and a positive and statistically significant effect in 2020. In both cases, the sign of the RD estimate refutes a story of strategic substitutes. That is, the data do not show that countries found it in their interest to do more when the U.S. signaled it would do less by electing Trump, nor would other countries do less when the U.S. elected Biden. When an anti-climate leader was elected in 2016, other countries followed suit and enacted fewer climate laws. In 2020, with the emergence of a pro-climate U.S. presidential winner, the international community responded with more climate action.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.009 (0.017)	-0.020 (0.019)	0.067*** (0.022)	0.070*** (0.020)
DV	Count	Binary	Count	Binary
DV Mean	0.094	0.109	0.105	0.116
Bandwidth	809.875	781.846	1616.020	1504.237
Effective Observations	10547	10149	20497	19701

Table A.15: RD Estimates of U.S. Presidential Elections on Climate Law Adoption

Results are robust to the inclusion of month fixed effects—which account for the influence of factors like the signing of new international environmental treaties, the onset of a global pandemic, or secular increases in worsening climate that may demand more action—as well as alternative specifications of the running variable, alternative bandwidths, placebo cutoffs, and sorting.

Table A.16 estimates the RD effect of U.S. elections without country fixed effects.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.009 (0.017)	-0.020 (0.019)	0.067*** (0.022)	0.070*** (0.020)
DV	Count	Binary	Count	Binary
Bandwidth (days)	810.142	782.037	1616.285	1504.298
Effective Observations	10547	10149	20497	19701

Table A.16: RD Estimates without Country Fixed Effects

The RD specification in the main text does not adjust for any other temporal shocks. Table A.17 controls for month fixed effects. These terms capture factors like signing of international agreements, the onset of global pandemic, or other time shocks that correlated with the adoption of climate laws around the election.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.014 (0.017)	-0.025 (0.019)	0.065*** (0.022)	0.066*** (0.020)
DV	Count	Binary	Count	Binary
Country fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Bandwidth (days)	829.619	824.242	1610.132	1510.144
Effective Observations	10746	10746	20298	19701

Table A.17: RD Estimates with Month Fixed Effects

Table A.18 shows that the main results hold if we define the running variable as months until the election instead of days.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.005 (0.016)	-0.015 (0.018)	0.073*** (0.024)	0.078*** (0.022)
DV	Count	Binary	Count	Binary
DV Mean	0.091	0.105	0.117	0.129
Bandwidth	30.714	31.677	51.471	47.863
Effective Observations	12139	12537	17711	16915

Table A.18: RD Estimates with Monthly Running Variable

Figures A.2 and A.3 serve as placebo tests for the RD effects that vary the cutoff in the running variable. I look at the 90 day period around each election. The estimate highlighted in red is the actual estimate.

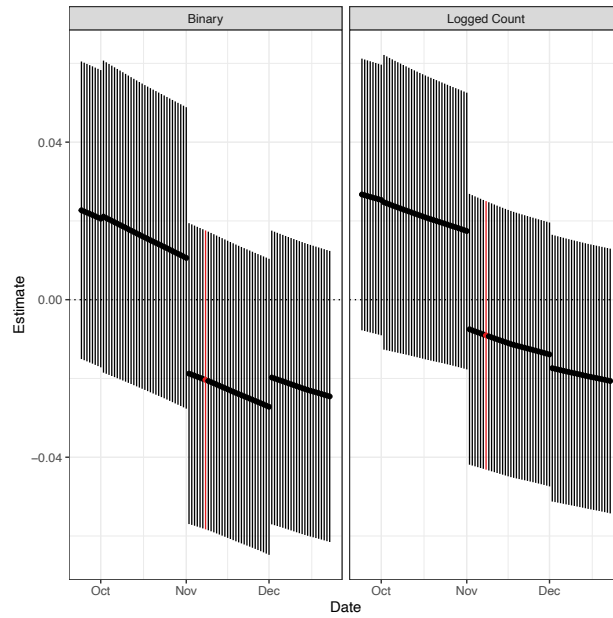


Figure A.2: RD Placebo Test (Trump)

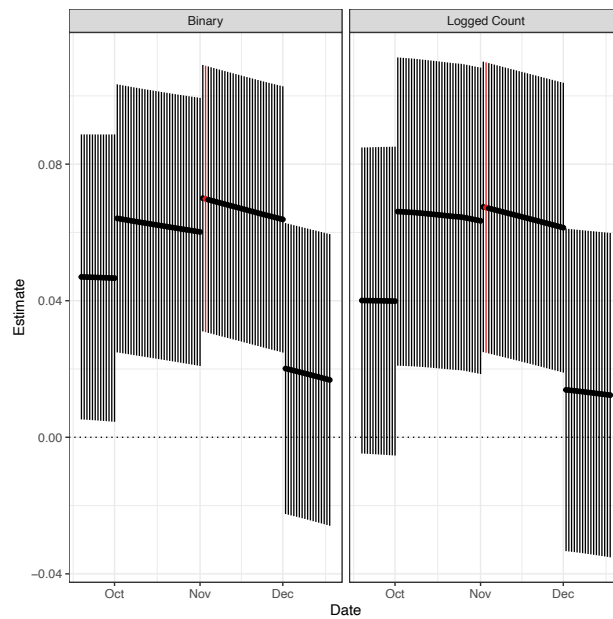


Figure A.3: RD Placebo Test (Biden)

Table A.19 shows the results of two tests of the no-sorting assumption, one based on McCrary (2008) and another based on Cattaneo et al. (2018). In both tests, the null hypothesis is that of no sorting. While the McCrary test is marginally significant for the Trump discontinuity, these results are suggestive of a lack of sorting in climate laws before/after the two U.S. elections.

Test	Election	p -value
McCrary (2008)	2016	0.091
McCrary (2008)	2020	0.149
Cattaneo et al. (2018)	2016	0.985
Cattaneo et al. (2018)	2020	0.931

Table A.19: RD Sorting Checks

Figure A.4 also shows how RD estimates are robust to different bandwidths. The estimate highlighted in red is the actual estimate. The Biden estimates are clearly robust; some of the Trump estimates change sign with very small bandwidths but they are not statistically significant.

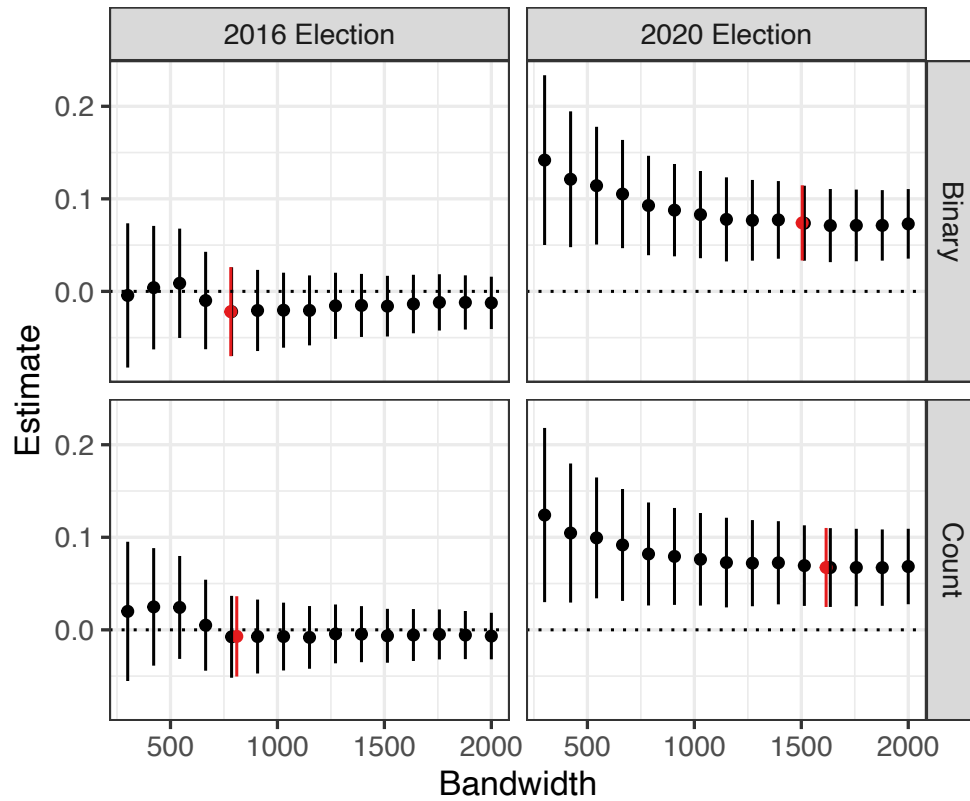


Figure A.4: RD Estimates with Alternative Bandwidths

C LLM-Based Classification of Learning Citations

This appendix describes how I construct and validate the two text-based measures of learning. The procedure has three stages—candidate generation, classification by a large language model (LLM), and validation against hand-coding—each detailed below. The full classifier prompt is reproduced at the end of the appendix.

Beginning from the full-text corpus of the Climate Change Laws of the World project and Climate Policy Radar (Nachmany et al. 2017; Institute 2025), restricted to national laws and policies in English,³ I scan every text block for mentions of another country using an exhaustive list of country names and case-sensitive acronyms with word-boundary matching. I discard self-references, boilerplate transposition of European Union and European Economic Area law, and substring artifacts. This yields 108,782 candidate references—passages in one country’s climate law that name another country—each retained together with its surrounding textual context.

I classify each candidate with a large language model. The primary classifier is Anthropic’s Claude Sonnet (model `claude-sonnet-5`), accessed through the Message Batches API on July 21 2026, with extended reasoning disabled and the default sampling temperature. Following common practice, I instruct the model to act as “an expert coder of climate-policy documents,” present it with the flagged passage and its surrounding context, and require it to return a single structured JSON object recording the three labels below.

Each candidate is coded on three nested indicators. The first, a *policy reference*, records whether the passage genuinely refers to the cited country’s own mitigation or clean energy policy, program, target, or outcome; it excludes mentions in which the country figures only as a trade or energy partner, a funder or donor, a bibliographic string, a currency unit, a geopolitical actor, or one entry in a descriptive roster. The second, a *learning citation*—the text’s first measure of learning—records whether the citing government substantively engages that policy or experience as an input to its own policymaking—describing its design or results, benchmarking against it, or drawing an explicit lesson from it. The third, an *economic value citation*, records whether a learning citation’s economic content—jobs, investment, cost, market or technological leadership, or a finance instrument—is stated in the passage itself or can be inferred. The domain is restricted throughout to climate mitigation and the clean energy transition; adaptation and other domains are excluded.

C.1 Validation

I follow standard practice for validating an LLM annotator against expert human coding (Gilardi, Alizadeh and Kubli 2023; Törnberg 2024). The codebook/prompt was developed iteratively on two initial samples and then frozen. Reliability is estimated on a fresh, held-out sample of 320 candidates (an initial 120 plus a further 200). Because instances of learning are relatively rare, the sample is stratified to contain enough positive cases (it holds 81

³Non-English documents are machine-translated by the data provider; I analyze the English-language text.

policy references, 38 learning citations, and 20 economic value citations), and all estimates are reweighted to corpus prevalence. I measure agreement with Cohen’s κ , together with precision and recall, and attach stratified-bootstrap 95% confidence intervals.

To guard against idiosyncrasies of any single model, I re-code the entire validation sample with a second, independently developed model family, OpenAI’s GPT-5 (model `gpt-5`), under the identical prompt and coding scheme. Agreement between the two families is reported in Table A.21.

Table A.20 reports the results for both model families.⁴ Agreement is moderate on every tier and, importantly, consistent across the two independently developed models. The economic value tier—the measure closest to the theory—is high-precision and moderate-recall for the classifier used at scale (precision 0.77, recall 0.48): it reliably flags explicit economic value citations but misses cases in which the economic content sits only in the surrounding context. The economic value counts reported in the main text should therefore be read as conservative lower bounds. Raw, non-chance-corrected agreement between the classifier and the coder is considerably higher—85%, 92%, and 96% on the three tiers—but κ is the appropriate metric given the rarity of the positive classes, since it removes the agreement expected by chance.

Tier	Claude (<code>claude-sonnet-5</code>)			GPT-5
	κ [95% CI]	Precision	Recall	κ [95% CI]
Policy reference	0.62 [0.51, 0.71]	0.70	0.74	0.58 [0.48, 0.68]
Learning	0.58 [0.42, 0.72]	0.69	0.58	0.57 [0.42, 0.69]
Economic value	0.57 [0.34, 0.75]	0.77	0.48	0.36 [0.15, 0.54]

Table A.20: Agreement with expert coding on the initial sample

Two further checks support the measure. First, the two model families agree with each other at least as closely as either agrees with the human coder—on 88%, 93%, and 96% of labels across the three tiers (Table A.21)—indicating that the coding scheme is not an artifact of a single model.⁵ Second, the classifier’s agreement with the human coding is insensitive to implementation choices: varying whether extended reasoning is enabled and whether requests are batched changes κ only at the margin (Table A.22). Where the classifier and coder disagreed, disagreements were re-adjudicated against the frozen codebook with full context. Applied to the full corpus, the classifier labels 1,913 learning citations and 790 economic value citations, the counts analyzed in the main text.

⁴Table shows Cohen’s κ , precision, and recall against a single expert coder on a held-out, stratified sample of 320 candidates (81 policy reference, 38 learning, 20 economic value positives), reweighted to corpus prevalence; 95% intervals from a stratified bootstrap. Both families reach moderate agreement on every tier; Claude is stronger on economic value citations.

⁵Table A.22 shows Cohen’s κ against the expert coding on the 120-candidate development sample; the adopted configuration’s agreement on the full $n = 320$ sample is reported in Table A.20. The choice of configuration barely affects agreement.

Tier	Raw agreement	Cohen’s κ [95% CI]
Policy reference	88%	0.70 [0.61, 0.78]
Learning	93%	0.64 [0.48, 0.76]
Economic value	96%	0.61 [0.38, 0.79]

Table A.21: Cross-model agreement (Claude vs. GPT-5)

Configuration	Policy reference	Learning	Economic value
Sonnet, extended reasoning on	0.54	0.48	0.55
Sonnet, extended reasoning off	0.58	0.43	0.68
Sonnet, batched, reasoning off (adopted)	0.56	0.46	0.73

Table A.22: Robustness to classifier configuration

Three limitations bear on interpretation. First, reliability is moderate ($\kappa \approx 0.57$ – 0.62 across tiers); economic value citations, though the most theory-relevant class, are the hardest to code, so moderate agreement is expected rather than a defect. Second, because recall for economic value citations is roughly 0.48, the measure is a conservative lower bound, and counts of levels of learning are best read as “at least.” Third, validation rests on a single expert coder—mitigated by the cross-model check and full-context adjudication.

C.2 Classifier Prompt

The model received the following instructions, codebook, and boundary examples, appended to each candidate passage (with the passage’s citing country, flagged country, document title and year, sentence, and surrounding context filling the bracketed fields).

Two nested tiers. Tier 1 (`learning_citation`): does country i engage country j ’s climate/energy policy or experience as something to *learn from*? Tier 2 (`economic_value_citation`, the paper’s θ): does that learning citation bear on the *economic value or feasibility* of j ’s green transition—whether stated explicitly or implied by an economic mechanism the passage invokes? Infer semantically; that is the reason to use a model here, not keyword matching. Nesting: `economic_value_citation` $\Rightarrow$ `learning_citation` $\Rightarrow$ `references_country_policy`.

Instructions. You are an expert coder of climate-policy documents. You are shown one passage from country {citing_country}’s national climate/energy law or policy. A regex flagged that it mentions {cited_country}. Judge only the reference to {cited_country}. Read the whole *Context*, not just the *Sentence*: the country name is often in one sentence (a membership list, a passing mention) while the learning or economic signal sits in an adjacent one. Decide from the full context, never from the flagged sentence alone. Return only a JSON object with these fields:

- `references_country_policy` (bool): a genuine reference to {cited_country}’s

own climate/energy policy, program, target, experience, or outcome? False if the country appears only as a trade or energy partner, transit point, war or sanctions actor, funder or donor, bibliographic string, currency unit, name collision, or one entry in a descriptive or enumerative list with no engagement of its policy. A list explicitly framed as learning sources (“learn from / examples such as A, B, C”) does count, for each named country.

- **country_actually_referenced** (ISO3 or “MULTIPLE”).
- **direction** (“i_learns_from_j” | “j_learns_from_i” | “peer/none”).
- **function**: **learning** · **compliance** (legal transposition/alignment with *j*’s or the EU rulebook) · **cooperation** (joint project, no lesson) · **trade_energy** · **geopolitics** · **descriptive_list** · **spurious**.
- **domain** (“mitigation” | “adaptation” | “both” | “other”). Only mitigation, clean energy, and the green transition are in scope; adaptation or other $\Rightarrow$ **learning_citation** = false.
- **learning_citation** (bool): true if **references_country_policy**, **direction** = **i_learns_from_j**, and the passage substantively engages the content of *j*’s policy, program, or its experience/effects as input to *i*’s own policy discussion. Explicit lesson-drawing language is sufficient but not required; describing *j*’s instrument, design, or results, or benchmarking against *j*, all count. False for out-of-scope domains; for compliance or accession obligations (“in line with the EU’s vision/acquis/directive” *alone*); for incidental data about *j* not tied to a policy; and for trade, energy supply, geopolitics, *j* as funder/donor, reverse direction, or a bare enumerative list.
- **economic_value_citation** (bool): true if **learning_citation** and the economic content is present in the passage itself. Counts: an explicit jobs, GDP, investment, or cost figure or claim; a stated “market/technology leader, champion, or pioneer”; a stated competitiveness or cost-saving gain; a named experience curve or documented deployment trajectory used as a feasibility model; deployment framed as investment; or a finance instrument whose economic function is the point. Does not count if the only economic angle is a generic chain the coder supplies (“efficiency $\Rightarrow$ saves money,” “any target $\Rightarrow$ growth”), emissions facts with no economic content, or bare “accelerate deployment.” Test: if you cannot quote the specific words carrying the economic claim, it is Tier 1, not Tier 2.
- **econ_reasoning** (string): if **economic_value_citation**, name the economic mechanism inferred, so the inference is auditable.
- **evidence_quote** (string): the ≤ 25 -word span justifying the labels.
- **confidence** (“high” | “med” | “low”).

D Documenting Learning: Ancillary Analysis

Table A.23 summarizes the corpus and the two learning measures. Of 108,782 candidate references across 6,720 national laws and policies, the classifier identifies 1,913 learning citations in 629 documents and, within them, 790 economic-value citations in 337 documents. The learning network spans 138 citing countries drawing on 117 distinct sources across 1,065 directed country pairs.

Quantity	Value
Candidate references (country mentions)	108,782
Total national laws and policies	6,720
Time span	1991–2025
Learning citations	1,913
in documents	629
Economic value citations	790
in documents	337
Citing countries	138
Cited sources	117
Directed country pairs	1,065

Table A.23: Summary statistics for the learning measure

Table A.24 ranks countries by the two roles that Figure 2 plots against one another: how often a country engages the experience of others, and how often its own experience is cited in return. The most active learners are middle-income and smaller advanced economies—Spain, Colombia, Japan, France, Turkey—while the most-cited sources are the early movers and large economies, led by the European Union, the United States, the United Kingdom, Germany, and China. Table A.25 lists the largest directed country pairs; ties to the European Union and among European neighbors dominate.

(a) Most active learners				(b) Most-cited sources		
Country	Cites	Countries Cited	Cites/Doc	Source	Cited	Citers
Spain	86	28	0.60	European Union	200	58
Colombia	84	32	1.14	United States	154	62
Japan	81	22	0.74	United Kingdom	115	51
France	66	15	0.41	Germany	112	45
Turkey	65	27	0.97	China	100	49
South Korea	61	17	0.67	France	83	37
India	48	18	0.50	Denmark	74	38
Norway	46	17	0.51	Japan	71	36
South Africa	44	30	0.75	Netherlands	69	39
Czechia	42	16	1.45	Sweden	61	28
United States	40	13	0.44	Norway	52	32
United Kingdom	36	16	0.23	India	50	33
Netherlands	36	14	0.44	Canada	46	23
Mexico	35	20	0.34	Brazil	37	27
Malaysia	34	15	1.48	Australia	36	25

Table A.24: Most active learners and most-cited sources (top 15)

Citing	Cited	Documents	with Economic value citations
Turkey	European Union	19	17
France	Netherlands	13	6
Japan	United States	13	9
South Korea	European Union	13	6
Colombia	European Union	11	6
Spain	France	11	6
United Kingdom	European Union	11	8
India	United States	10	5
Japan	European Union	10	9
Montenegro	European Union	9	4
Norway	Sweden	9	1
South Korea	Japan	9	5
Canada	United States	8	5
Croatia	Slovenia	8	5
France	Germany	8	4

Table A.25: Largest directed country pairs (top 15).

Table A.26 ranks countries by economic value citations alone. The set of countries invoking economic benefits in their citations is not identical to the set of learners overall: Japan and France, in particular, rise to the top, reflecting a greater tendency to frame others' policies in economic terms. On the receiving side, the sources are the large economies whose green transitions carry the clearest economic signal—the European Union, the United States, Germany, and China.

(a) Cite the economic value of others		(b) Cited for economic value	
Country	Economic value cites	Cites/Doc Source	Economic value cites
Japan	51	0.46 European Union	106
France	44	0.27 United States	72
Colombia	42	0.57 Germany	53
Spain	37	0.26 China	50
Türkiye	35	0.52 United Kingdom	47
Mexico	25	0.25 France	33
South Korea	24	0.26 Denmark	27
South Africa	24	0.41 Norway	27
India	21	0.22 Japan	24
Norway	20	0.22 Netherlands	24
United Kingdom	17	0.11 South Korea	20
Croatia	17	0.37 Brazil	18

Table A.26: Economic value citations, by country (top 12)

About a third of the corpus is machine-translated by the data provider, raising the concern that translation degrades the signal. Table A.27 splits the learning citations by whether the citing document was translated. Learning appears in both strata; the per-document rate is somewhat higher in English-original text (0.66 versus 0.46). This gap is consistent with some loss of signal in translation, which would make the measure conservative for non-English documents.

Table A.27: Learning citations by document language.

Documents	Learning	Economic value	Learning / Doc
English-original	1,233	501	0.66
Machine-translated	680	289	0.46

We may be worried that policy citations that invoke economic benefits of decarbonization could reflect international competition rather than policy learning. I manually reviewed the 790 economic benefit citations and found that only 20 of them, or 2.5%, explicitly mention international competition. Table A.28 provides examples from this small subset.

Citing Country	Cited Country	Document (year)	Excerpt
Norway	Singapore	Better Growth, Lower Emissions Strategy (2017)	“A number of countries are now developing policies to promote green competitiveness. In Singapore, the authorities are pursuing an active policy to develop, create markets for and deploy their own green technology. Canada is seeking to diversify its economy, and has set clear goals for its green growth policy to result in cuts in greenhouse gas emissions, more jobs and economic growth.”
India	United States	National Mission for Electric Mobility Plan 2020 (2012)	“In order to be globally competitive, it is essential for the Indian companies to reach capability levels comparable to US, Japan, China and Korea. Globally, Japan, United States and Korea have strong capabilities in lithium batteries. Japan leads with 5500 patents for battery cells and BMS and Korea and US follow suit with 2500 2000 and patents respectively.”
United States	China	Federal Sustainability Plan: Catalyzing America’s Clean Energy Industries and Jobs (2021)	“The European Union, the United Kingdom, and Canada have prepared their domestic industries and workers for the energy transition through industrial policy strategies. Furthermore, China’s domestic industry support has made it the global clean energy manufacturing leader. Moreover, the United States has become reliant on other countries for procuring clean energy technologies of the future, including zero-emission vehicles and certain renewable energy technologies.”

Table A.28: Citations Invoking Economic Benefits: International Competition

E Empirical Implications: Robustness and Ancillary Analysis

This appendix presents robustness and ancillary specifications for the mass beliefs analysis in the main text.

E.1 Mass Beliefs

Statistic	N	Mean	St. Dev.	Min	Max
Belief CC Serious	390	24.050	10.808	8.474	52.239
Own Laws	390	0.954	1.005	0.000	3.996
Other Laws	390	2.176	0.961	0.480	4.539

Table A.29: Summary Statistics for Mass Beliefs

Tables A.30-A.32 demonstrate that the results presented in the main text are robust to alternative specifications, namely the inclusion of a lagged dependent variable, the exclusion of time effects (either year fixed effects or country-year trends), and the use of a recoded 1-4 survey scale.

Implication	Regressor	Outcome	Estimate (SE)	<i>N</i>
1	Own Seriousness	log(Own Laws)	0.129 (0.096)	210
2	log(Other Laws)	Own Seriousness	0.536*** (0.122)	362
3	Other Seriousness	log(Own Laws)	0.294*** (0.090)	212

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.30: Empirical Implications of the Learning Mechanism (including Lagged DVs)

Implication	Regressor	Outcome	Estimate (SE)	<i>N</i>
1	Own Seriousness	log(Own Laws)	0.287*** (0.061)	263
2	log(Other Laws)	Own Seriousness	1.114*** (0.204)	390
3	Other Seriousness	log(Own Laws)	0.284*** (0.050)	265

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.31: Empirical Implications of the Learning Mechanism (without Time Effects)

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	log(Own Laws)	0.151* (0.074)	263
2	log(Other Laws)	Own Seriousness	0.663*** (0.159)	390
3	Other Seriousness	log(Own Laws)	0.266*** (0.074)	265

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.32: Empirical Implications of the Learning Mechanism (Recoded 1-4 Scale)

The main text measures effort with the count of laws; Tables A.33 and A.34 use the CAPMF (Nachtigall et al. 2024) and EPS (Botta and Kořluk 2014; Kruse et al. 2022) measures of policy stringency instead. The results are weaker and generally fall short of conventional levels of statistical significance. However, there is a negative and significant relationship between other nations' seriousness and one's own policy stringency if stringency is measured using the CAPMF. As this is a substantively demanding test, it may be reflective of the other strategic forces like free-riding at play.

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	Own Stringency	−0.004 (0.020)	322
2	Avg. Other Stringency	Own Seriousness	−0.305 (0.199)	390
3	Other Seriousness	Own Stringency	−0.138*** (0.032)	324

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.33: Empirical Implications of the Learning Mechanism (CAPMF)

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	Stringency	6.6×10^{-5} (0.167)	189
2	Avg. Other Stringency	Own Seriousness	−0.036 (0.066)	306
3	Other Seriousness	Stringency	0.160 (0.165)	189

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.34: Empirical Implications of the Learning Mechanism (EPS)

F Alternative Model with $n > 2$ Countries

The model in the main text assumes a two-country interaction. The theoretical purchase of this approach was that it allowed for a parsimonious study of forward-looking incentives to exert effort or not. In this section, I describe a model that features $n > 2$ countries but remove this strategic interdependence. Similar to observational learning models (Banerjee 1992; Bikhchandani, Hirshleifer and Welch 1992; Smith and Sørensen 2000), countries are only backward-looking in their valuation of contributions to global effort.

F.1 Model Setup

Consider sequential climate policymaking between n countries indexed by $i = 1, \dots, n$ who decide whether to pursue climate reforms $a_i = 1$ or not $a_i = 0$. The action $a_i = 1$ represents climate reforms or policies instituted to facilitate a green transition in country i , while $a_i = 0$ represents maintaining the status quo. These countries take actions in a fixed order and can observe the choices of all countries before them.

Countries' payoffs to climate reform depend on two uncertain elements: the global benefit to a green transition and the private domestic cost of implementing reforms. The global benefit to a green transition is a binary state variable $\theta \in \{0, 1\}$. No country knows the true realization of θ —whether or not the green transition pays off or will be “successful” is unknown—but share the common prior $P(\theta = 1) = \pi \in (0, 1)$. If a country does not take climate action, it receives a payoff of zero. By taking climate action, country i receives a benefit normalized to 1 only if $\theta = 1$: this captures the idea that countries only want to pursue climate reforms if it is appropriate to do so or if the green transition is sufficiently likely to be successful.

While the benefits of a green transition are state dependent, their costs are not: choosing $a_i = 1$ comes at a cost $c_i \sim U[0, 1]$. These costs represent the domestic political feasibility of the green transition. Country i 's costs of implementation are privately known, drawn independently for each country, and are independent of θ .

There are two information sources that countries have at their disposal when determining whether to implement climate policy. The first is the history of observed actions, $h_i = (a_1, \dots, a_{i-1})$. Countries learn about the suitability of green policy θ through the behavior of others. They also receive conditionally independent private signals, which, along with the prior, generate private beliefs $p_i \in [0, 1]$. I work with these beliefs rather than signals and the prior π directly (p_i is a sufficient statistic). These beliefs are not publicly known, but insofar as they translate into actions they may partially inferred. Let the cumulative distribution function of a private belief p in state θ be $F(p|\theta)$ with density $f(p|\theta)$ such that $F(p|1) < F(p|0)$.⁶

⁶By Bayes's Rule, the state-conditional densities $f(p|\theta)$ satisfy $p = \frac{\pi f(p|1)}{f(p)}$ and $1 - p = \frac{(1-\pi)f(p|0)}{f(p)}$ with $f(p) = \pi f(p|1) + (1-\pi)f(p|0)$. Then $\frac{f(p|1)}{f(p|0)} = \frac{p}{1-p} \frac{1-\pi}{\pi}$; this stochastic ordering implies that the conditional distributions are mutually absolutely continuous, share the same support, and that $F(p|1) < F(p|0)$ for all private beliefs strictly inside the support.

Given some history of countries' climate policies h_i , define the public belief $P(\theta = 1|h_i)$ as the informational content about the suitability of the green transition. There is an associated public likelihood ratio $\ell_i = \frac{1-P(\theta=1|h_i)}{P(\theta=1|h_i)}$ such that lower values of ℓ_i imply a greater likelihood that $\theta = 1$, or that a green transition would be successful.

The climate problem is often described as one of strategic substitutes because it is nationally costly to exert effort to address climate change despite that this effort provides a global benefit. To capture this tension, I introduce *collective action penalties*, which are action-specific, history-dependent costs (Eyster et al. 2014). It becomes more costly to pursue climate reform if many other countries have done so already. In reduced form, these penalties capture the strategic substitutability of climate actions across countries present in other models. Denote collective actions penalties as $z(h_i)$ (and suppress dependence on h_i where it is not confusing). This function is increasing in the number of countries who have already taken climate action, or $\sum h_i$. For simplicity, assume the penalty is bounded, $z_i \in [0, 1]$; since the domestic costs c_i are on the same scale, there is no explicit assumption as to whether implementation costs are greater than collective action penalties. As an example, consider a linearly proportional cost function for any country $i \geq 2$ (with $z_1 = 0$),

$$z_i = \frac{\sum h_i}{i-1}.$$

In countries' payoffs I scale these externalities by $k > 0$ in order to parameterize the extent to which countries weigh potential complementarities (generated by information about θ) and potential substitutes (generated by collective action penalties). This parameter can be thought of as scaling the extent to which countries internalize free-riding concerns; larger k implies stronger free-riding incentives as collective action penalties are weighted more heavily.

Given this setup, country i 's payoff can be written as

$$u_i(a_i, h_i, \theta; c_i) = a_i(\theta - c_i - kz(h_i)).$$

A strategy for country i is a choice to implement climate reforms or not, $a_i \in \{0, 1\}$, given the choices of other prior-moving countries contained in history h_i , and its type (p_i, c_i) , comprised of its private belief about θ and its domestic costs of implementing green policy. I examine weak perfect Bayesian equilibria and derive all posterior beliefs via Bayes's Rule.

F.2 Results and Proofs

Fix a history $h_i = (a_1, \dots, a_{i-1})$ of past climate policy adoption decisions that induce a public likelihood ratio ℓ_i and potential collective action penalties z_i . Given ℓ_i and the private belief p_i , country i can make an assessment about the appropriateness of climate policy, i 's posterior belief that $\theta = 1$ is defined as

$$\mu_i = P(\theta = 1|p_i, \ell_i) = \frac{p_i}{p_i + (1 - p_i)\ell_i}.$$

Then, country i prefers to implement climate reforms if and only if

$$\begin{aligned}
\mu_i - c_i - kz_i &\geq 0 \\
\Leftrightarrow \mu_i &\geq c_i + kz_i \\
\Leftrightarrow p_i &\geq \frac{\ell_i(c_i + kz_i)}{1 - (c_i + kz_i) + \ell_i(c_i + kz_i)} \equiv \tilde{p}(c_i, \ell_i).
\end{aligned}$$

Country i pursues climate action if and only if their private belief about a successful green transition is sufficiently high, given the domestic political costs of implementing climate reforms and the potential collective action penalties. If $\tilde{p}(c_i, \ell_i) > 1$, then i never takes climate action regardless of the value of p_i , which occurs whenever $c_i > 1 - kz_i \equiv \bar{c}_i$. Intuitively, if the domestic costs of implementing climate policy are prohibitively high, it does not matter how optimistic i is about the green transition, implementing green policy is not domestically feasible. Then, for any $c_i \in [0, \bar{c}_i]$, i pursues climate action if and only if $p_i > \tilde{p}(c_i, \ell_i)$, which occurs with probability $1 - F(\tilde{p}(c_i, \ell_i)|\theta)$.

Lemma A.1. *The threshold $\tilde{p}(c_i, \ell_i)$ is:*

- *increasing in the public likelihood ratio ℓ_i ;*
- *increasing in domestic implementation costs c_i ;*
- *increasing in collective action penalties z_i ;*
- *increasing in the strength of free-riding incentives k .*

Proof of Lemma A.1.

$$\begin{aligned}
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial \ell_i} &= \frac{(1 - c_i - kz_i)(c_i + kz_i)}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial c_i} &= \frac{\ell_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} &= \frac{k\ell_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial k} &= \frac{\ell_i z_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0.
\end{aligned}$$

□

Proposition A.1. *Let $\alpha^*(a_i|\ell_i, \theta)$ be the probability that country i takes climate action a_i in state θ . Then*

$$\alpha^*(1|\ell_i, \theta) = \int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|\theta) dc_i = 1 - \alpha^*(0|\ell_i, \theta).$$

Proof of Proposition A.1. Immediate from text. $\square$

Corollary A.1. *Climate policy is informative about θ . The probability of climate action is greater when $\theta = 1$ versus $\theta = 0$: $\alpha^*(1|\ell_i, 1) > \alpha^*(1|\ell_i, 0)$. The probability of climate inaction is greater when $\theta = 0$ versus $\theta = 1$: $\alpha^*(0|\ell_i, 1) < \alpha^*(0|\ell_i, 0)$.*

Proof of Corollary A.1.

$$\begin{aligned}\alpha^*(1|\ell_i, 1) - \alpha^*(1|\ell_i, 0) &= \left(\int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|1) \, dc_i \right) - \left(\int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|0) \, dc_i \right) \\ &= \int_0^{\bar{c}_i} F(\tilde{p}(c_i, \ell_i)|0) - F(\tilde{p}(c_i, \ell_i)|1) \, dc_i > 0,\end{aligned}$$

where the result follows from the stochastic ordering of private beliefs. $\square$

Corollary A.2. *The probability of climate action is increasing in the public optimism about a successful green transition, $\frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} \leq 0$.*

Proof of Corollary A.2. Differentiating with respect to ℓ_i yields

$$\frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} = - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}_i}{\partial \ell_i} \, dc_i \leq 0.$$

$\square$

Corollary A.3. *The probability of climate action is decreasing in collective action penalties, $\frac{d\alpha^*(1|\ell_i, \theta)}{dz_i} \leq 0$.*

Proof of Corollary A.3. By the Leibniz integral rule, differentiating with respect to z_i yields

$$\begin{aligned}\frac{d\alpha^*(1|\ell_i, \theta)}{dz_i} &= \frac{\partial \bar{c}_i}{\partial z_i} - F(\tilde{p}(\bar{c}_i, \ell_i)|\theta) \frac{\partial \bar{c}_i}{\partial z_i} - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} \, dc_i \\ &= - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} \, dc_i \leq 0.\end{aligned}$$

where the first two terms simplify because $\tilde{p}(\bar{c}_i, \ell_i) = 1$. $\square$

Corollary A.2 states that more optimistic public beliefs about a successful green transition begets more climate action. That is, these beliefs endogenously generate *complementarities* in countries' climate actions. Conversely, increased collective action penalties—which arise because more countries have already pursued climate policies—depress subsequent climate action, as stated in Corollary A.3. The actions of prior movers induce *substitution* in the behavior of later policymakers.

Which factor dominates? Under what conditions are countries' actions strategic complements or strategic substitutes in equilibrium? To conceptualize this, I consider the ratio of the marginal effects of public beliefs and collective action penalties. Define $\rho(\ell_i, z_i|\theta)$ as

$$\rho(\ell_i, z_i|\theta) = \frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} \bigg/ \frac{d\alpha^*(1|\ell_i, \theta)}{dz_i}.$$

The magnitude of $\rho(\ell_i, z_i|\theta)$ is always positive, but we can think about which factor dominates—strategic complementarities that stem from increased public beliefs or strategic substitutes from collective action penalties—based on where it is greater than or less than 1. If $\rho(\ell_i, z_i|\theta) > 1$, then, all else equal, varying public beliefs has a larger effect on the equilibrium probability of climate action than does varying collective action penalties. In this case, we can say that the net effect of other countries' behavior generates complementarities for country i . By contrast, when $\rho(\ell_i, z_i|\theta) < 1$, then the incentives to free ride swamp the potential benefits from climate policy investment.

Proposition A.2. *Complementarity effects dominate when free-riding incentives are small, and substitution effects dominate when free-riding incentives are large: there exists a threshold $\bar{k}_i$ such that if $k < \bar{k}_i$ then $\rho(\ell_i, z_i|\theta) > 1$.*

Proof of Proposition A.2. It follows that

$$\rho(\ell_i, z_i|\theta) > 1 \Leftrightarrow \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}_i}{\partial \ell_i} dc_i > \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} dc_i.$$

Define $Q(k) = (1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2 \geq 0$, which is the denominator of the comparative statics on $\tilde{p}(c_i, \ell_i)$. Simplifying yields

$$\int_0^{\bar{c}_i} \frac{f(\tilde{p}(c_i, \ell_i)|\theta)}{Q(k)} \left((1 - c_i - k_i)(c_i + kz_i) - k\ell_i \right) dc_i > 0.$$

Now note that for any $k < \frac{1}{z_i}$, the integral is well-defined (otherwise $\bar{c}_i = 0$). Furthermore, for any $k < \frac{1}{z_i}$, $\frac{f(\tilde{p}(c_i, \ell_i)|\theta)}{Q(k)} \geq 0$ and we are integrating over a positive interval of the c_i space. So the integrand is negative if and only if

$$(1 - c_i - k_i)(c_i + kz_i) - k\ell_i < 0,$$

which simplifies to $k > \frac{z_i - 2c_i z_i - \ell_i + \sqrt{\ell_i^2 - 2\ell_i z_i + 4c_i \ell_i z_i + z_i^2}}{2z_i^2} \equiv \bar{k}_i$. Hence a sufficient condition for the integrand to be negative is if $k > \bar{k}_i$ which implies that $\rho(\ell_i, z_i|\theta) < 1$. $\square$

We can now use the model to think about the long-run dynamics of climate policy across countries based on the analysis in the previous subsection. We have shown that the decision

problem facing each country at the time of climate adoption is static, meaning history-relevant parameters such as ℓ_i and z_i can be treated in reduced form, but now wish to trace the evolution of actions and beliefs across countries.

Since private signals are conditionally independent, the likelihood ratio updates such that

$$\ell_{i+1} = \varphi(a_i, \ell_i) = \ell_i \frac{\alpha^*(a_i|\ell_i, 0)}{\alpha^*(a_i|\ell_i, 1)}.$$

Observe that by Corollary A.1, relative to ℓ_i , ℓ_{i+1} shrinks if $a_i = 1$ but ℓ_{i+1} grows if $a_i = 0$. The public belief becomes more or less optimistic depending on the previous action a_i , which in turn informs the decision to enact climate policy in the subsequent period. Then, given the updated public belief and any additional collective action penalties, country $i+1$ considers the tradeoff between implementing climate reforms and incurring domestic implementation costs and collective action penalties or free-riding, where climate policy occurs with probability $\alpha^*(1|\ell_{i+1}, \theta)$.

As is standard in the informational cascades and herding literature (e.g., [Smith and Sørensen 2000](#)), convergence results are stated conditioning on $\theta = 1$. This is also the more interesting case from a substantive perspective anyway, as this is where the tradeoff between the two mechanisms is present.

Lemma A.2. *Conditional on $\theta = 1$, the public likelihood ratio $\langle \ell_i \rangle$ is a martingale.*

Proof of Lemma A.2. Recall that the public likelihood ratio updates according to

$$\ell_{i+1} = \ell_i \frac{\alpha^*(a_i|\ell_i, 0)}{\alpha^*(a_i|\ell_i, 1)},$$

by the conditional independence of signals. Taking expectations yields

$$\begin{aligned} E[\ell_{i+1}|\ell_1, \dots, \ell_i, \theta = 1] &= \sum_{a \in \{0,1\}} \alpha^*(a|\ell_i, 1) \ell_i \frac{\alpha^*(a|\ell_i, 0)}{\alpha^*(a|\ell_i, 1)} \\ &= \ell_i \sum_{a \in \{0,1\}} \alpha^*(a|\ell_i, 0) \\ &= \ell_i. \end{aligned}$$

□

Proposition A.3. *In the limit, countries learn whether the green transition will be successful: public beliefs converge to the true state of the world almost surely.*

Proof of Proposition A.3. Without loss of generality condition on state $\theta = 1$. Since $\langle \ell_i \rangle$ is a martingale and all values are nonnegative, it converges almost surely to a random variable $\ell_\infty = \lim_{i \rightarrow \infty} \ell_i$ with support $[0, \infty)$ by the Martingale Convergence Theorem ([Doob 1953](#)).

This rules out nonstationary limit beliefs. Since private beliefs p_i are unbounded within $[0, 1]$, then the only stationary finite likelihood ratio in state $\theta = 1$ is 0, so $\ell_\infty \rightarrow 0$ almost surely (Smith and Sørensen 2000, Theorem 1). $\square$

Proposition A.4. *In the limit, countries take the correct action $a_i = \theta$ if and only if $c_i \leq \bar{c}_i$.*

Proof of Proposition A.4. Recall that i chooses $a_i = 0$ if $c_i > \bar{c}_i$ for any private belief p_i , which occurs with probability $P(c_i > \bar{c}_i) = kz_i$. This probability is increasing in z_i , which increases in the number of countries that choose $a_i = 1$. Now suppose that $\theta = 1$ and $c_i \leq \bar{c}_i$ so i chooses $a_i = 1$ iff $p_i \geq \tilde{p}(c_i, \ell_i)$. By Proposition A.3, the likelihood ratio converges almost surely to $\ell_i \rightarrow 0$. Then we have $\lim_{\ell_i \rightarrow 0} \tilde{p}(c_i, \ell_i) = 0$. Hence conditional on $c_i < \bar{c}_i$, i chooses $a_i = 1 = \theta$ for any private belief.

If countries are taking action on a measure zero subset, $\bar{c}_j \rightarrow 0$ or $k > \frac{1}{z_j}$ for some $j \leq n$, then countries pool on $a_j = 0 \forall j, \dots, n$. $\square$

Corollary A.4. *Let $z_i(h_i) = \frac{\sum_{i=1}^n h_i}{i-1}$. The probability of climate action converges to $\frac{1}{1+k}$.*

Proof of Corollary A.4. Conditional on state $\theta = 1$, climate action occurs with probability $\bar{c}_i$, as $\tilde{p}(c_i, \ell_i) \rightarrow 0$ as $\ell_i \rightarrow 0$. Moreover, given that $z_i = \frac{\sum_{i=1}^n h_i}{i-1}$, it is a linear proportional function of previous actions, so in the limit, $z_i \rightarrow \alpha^*(1|\ell_i, \theta)$. Then we have

$$\begin{aligned} \alpha^*(1|\ell_i, \theta) &= \bar{c}_i = 1 - kz_i \\ &= 1 - k\alpha^*(1|\ell_i, \theta) \\ \Leftrightarrow \alpha^*(1|\ell_i, \theta) &= \frac{1}{1+k}. \end{aligned}$$

$\square$

Corollary A.5. *In the limit, $\rho(\ell_i, z_i|\theta) > 1$ if $k < 1 - c_i$.*

Proof of Corollary A.5. Per Proposition A.2, a sufficient condition for $\rho(\ell_i, z_i|\theta) > 1$ is $k < \bar{k}_i$ where $\bar{k}_i = \frac{z_i - 2c_i z_i - \ell_i + \sqrt{\ell_i^2 - 2\ell_i z_i + 4c_i \ell_i z_i + z_i^2}}{2z_i^2}$. Then in the long run, $\lim_{\ell_i \rightarrow 0} \bar{k}_i = 1 - c_i$. $\square$

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